



University of Papua New Guinea

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Python

What is Python programming language?



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Python is a high-level, general-purpose, interpreted programming language.

- 1) High-level
- 2) General-purpose
- 3) Interpreted

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1) High-level

- Python is a high-level programming language that makes it easy to learn.
- Python doesn't require you to understand the details of the computer in order to develop programs efficiently.

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2) General-purpose

Python is a general-purpose language. It means that you can use Python in various domains including:

- Web applications
- Big data applications
- Security
- Data science, machine learning, and AI
- Desktop software
- Mobile apps



3) Interpreted

- Python is an interpreted language.
- To develop a Python program, you write Python code into a file called source code.
- To execute the source code, you need to convert it to the machine language that the computer can understand.
- And the Python **interpreter** turns the source code, line by line, once at a time, into the machine code when the Python program executes.

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Why Python?

- Python is quite easy to learn in comparison with other programming languages.
- Python syntax is clear and beautiful.
- Python has a large ecosystem that includes lots of libraries and frameworks.
- Python is cross-platform.
- Python programs can run on Windows, Linux, and macOS.

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Installation:

Download Anaconda – A package/a distribution of the Python and R programming languages for scientific computing, that aims to simplify package management and deployment.

The distribution includes data-science packages and other packages suitable for Windows, Linux, and macOS.

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Continuation of statements:

```
if (a == True) and (b == False) and \  
    (c == True):  
    print("Continuation of statements")
```

Variable in Python:

- Variable names can contain only letters, numbers, and underscores (_).
- They can start with a letter or an underscore (_), not with a number.

```
message = 'Hello, Python!'  
print(message)
```

```
message = 'Hello World!'  
print(message)
```

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Python Syntax

define main function to print out something

```
def main():
```

```
    i = 1
```

```
    max = 10
```

```
    while (i < max):
```

```
        print(i)
```

```
        i = i + 1
```

call function main

```
main()
```

Comments

```
# This is a single line comment in Python
```

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Python string:

- In Python, anything inside quotes is a string.
- You can use either single quotes or double-quotes.

```
message = 'string in Python'
```

```
message = "This is also a string in double quotes"
```

```
message = "It's a string"
```

```
message = ' "Beautiful is better than ugly.". Said Tim Peters'
```

```
message = 'It\'s also a valid string'
```

raw strings by adding the letter r before the first quote.

```
message = r'C:\python\bin'
```

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multiline strings in python:

```
help_message = '''
Usage: mysql command
-h hostname
-d database name
-u username
-p password
'''
```

```
print(help_message)
```

Python strings with the f-strings:

```
name = 'John'
message = 'Hi'
```

```
name = 'John'
message = f'Hi {name}'
print(message)
```

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Concatenating Python strings:

```
greeting = 'Good ' 'Morning!'
print(greeting)
```

```
greeting = 'Good '
time = 'Afternoon'
```

```
greeting = greeting + time + '!'
print(greeting)
```

Accessing string elements:

```
str = "Python String"
```

```
print(str[0])          # P
```

```
print(str[1])          # y
```

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```
str = "Python String"
    print(str[-1])      # g
    print(str[-2])      # n
```

Getting the length of a string:

```
str = "Python String"
str_len = len(str)
print(str_len)
```

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Python strings are immutable:

```
str = "Python String"
```

```
str[0] = 'J'
```

#It will through an error.

```
str = "Python String"
```

```
new_str = 'J' + str[1:]
```

```
print(new_str)
```

#Output

Jython String

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Python Numbers:

- Python supports integers, floats, and complex numbers.

Integers

- The integers are numbers such as -1, 0, 1, 2, 3, .. and they have type int.

Floats

- Any number with a decimal point is a floating-point number.

Complex number:

- A complex number is created from real numbers.
- Python complex number can be created either using direct assignment statement or by using `complex()` function.

- `complex([real[, imag]])`
 `c = 3 + 6j`
 `print(type(c))`
 <class 'complex'>